

STRATA: A Stratified Process-Mining Framework for HVAC Fault Detection and Diagnosis

Yassh Singh

Supervisors: Dr. Anthony Aighobahi, Dr. Mridula Sharma
Thompson Rivers University — UREAP 2026

September 2026

Abstract

Faults in building HVAC equipment persist because the tools meant to catch them either raise alarms that operators learn to ignore, or flag anomalies that nobody can act on. STRATA turns ordinary one-minute trend data into an event log along the control sequence, discovers process models at the device and unit level, and scores each day through eight calibrated, physically interpretable channels against a measured noise floor, so that every alarm names a location, states its evidence and is held to a false-alarm budget. It was built under a falsification protocol: every hypothesis was pre-registered with the observation that would refute it, every result was recomputed independently before it was quoted, and automated tests pin the published numbers and the falsifiers that fired.

On five public LBNL benchmark systems, including the two fan-powered terminal units for which no detection results had been published before, STRATA detects 158 of 175 seeded faults, most on the first day, at 26 false-alarm days in 480 held-out fault-free days (1 to 9.4% per system). Two systems were onboarded blind, by configuration alone. The alarm names the correct terminal unit in 37 of the 50 detections with known ground truth and a wrong one in none, and the faulted subsystem in 83% of all detections. The false-alarm budget holds under injected sensor error, change-of-value logging and schedule shifts (worst 7 of 96 days) and on one measured rooftop unit (3 of 49).

Process mining helps to get detection but is not detecting on its own. The strata and the event abstraction carry the localisation, and invariants that discovery surfaced became several of the rules that detect. Conformance checking of the discovered models, pre-registered as the detector, was refuted: over 199 scenarios on six systems, no fault is caught by the models alone. A principal-component baseline detects three fewer of the 73 scenarios it shares and cannot name the device. A third-party Guideline 36 rule battery, held to the same gate, matches STRATA within one detection on the air handler and detects a third as many faults on the terminal units, so the pre-registered improvement claim over current practice is withdrawn there. The evaluation also produced eight machine-verified errata in the benchmark datasets. Every number regenerates from the public repository.

1 Introduction

Buildings and construction account for 32% of global final energy use and 34% of energy-related carbon dioxide emissions [20]. HVAC accounts for up to half of an individual building’s consumption [3] and for about 30% of all commercial building energy in the United States [8], and 15–30% of that is wasted through faulty operation and poor control [3,11]; correcting the common faults saves 5–30% of whole-building energy [3,17]. The faults persist even where detection is installed. Across

60,000 pieces of equipment in 317 commercial buildings, Crowe and colleagues found 245 faults per building per month and 40% of air handlers in a faulted state on any given day, in buildings that run modern automation systems with alarms [6]. Seven of their ten most common faults are behaviour-based: wrong schedules, unreleased overrides, missed setpoint resets, broken economiser sequences. A threshold alarm or a pre-written rule [18] catches such a fault only if someone wrote that rule in advance [10]. The public benchmarks used in this paper seed component, sensor and control-loop faults and none of that class, so this argument motivates the design and is not tested by it.

Research has gone the other way. Machine learning has been surveyed across 211 HVAC fault-detection studies [3], with accuracies saturated at 97–99% on the standard benchmarks [16, 19] and interpretability added afterwards at the level of input features [5, 7, 12, 22, 23], which tells an operator that a sensor mattered, not what the building should have been doing. Of 65 primary studies audited in 2025, two shared code links, one of them broken, and 72% do not say whether their data is public [13].

What an operator needs is a detector that works from the trend data already recorded, names the unit and subsystem, explains the alarm in the physics of the equipment, is held to a false-alarm rate measured on the building’s own fault-free days, and reaches a new building by configuration rather than new code. Process mining, the one methodology built to model the sequence in which a system moves between states, has not been brought to this problem: to our knowledge no published work applies event-log process mining to building HVAC fault detection. The 211-study survey contains no process-mining entry [3], the reviews of process mining in Industry 4.0 and on sensor data contain no building HVAC application [1, 4], the Lawrence Berkeley National Laboratory taxonomy of fault-detection methods has no such leaf [9], and our own exact-phrase searches of the OpenAlex corpus return none [15].

STRATA applies process mining in stratified form: sensor streams are abstracted into events along ASHRAE Guideline 36 control sequences [2], Petri nets are discovered per device and per unit [14, 21], and the channels that read the log sit beside physics rules and residuals that read the raw signals, each calibrated on its own noise. Our first hypothesis, that the discovered net would itself be the detector, was pre-registered and refuted: a single hand-written rule out-covered the model channel where it was strongest. What settled is the division of labour, and it is a productive one. The strata and the event abstraction give every alarm its location, the frequency and oscillation channels are the only ones that catch the unstable-controller faults, invariants that discovery surfaced became several of the rules that detect, and conformance checking of the discovered nets, the part we expected to detect, contributes no detection of its own. Process mining helps to get detection but is not detecting on its own.

The paper contributes (1) the framework and the detector it yields, evaluated on five public benchmark systems including the two fan-powered terminal units for which no detection results had been published before; (2) what process mining contributes to HVAC fault detection, established under pre-registration: the first such application, whose structure gives the detector its location and several of its rules, and whose conformance channel catches no fault the calibrated channels miss; (3) an evaluation protocol of pre-registration, honoured falsifiers, independent recomputation and full artefact regeneration; and (4) eight machine-verified errata in the benchmark datasets.

References

- [1] Katsiaryna Akhramovich, Estefanía Serral, and Carlos Cetina. A systematic literature review on the application of process mining to Industry 4.0. *Knowledge and Information Systems*,

66(5):2699–2746, 2024.

- [2] ASHRAE. ASHRAE Guideline 36-2021: High-performance sequences of operation for HVAC systems. American Society of Heating, Refrigerating and Air-Conditioning Engineers, Peachtree Corners, GA, 2021.
- [3] Jian Bi, Hua Wang, Enbo Yan, Chuan Wang, Ke Yan, Liangliang Jiang, and Bin Yang. AI in HVAC fault detection and diagnosis: A systematic review. *Energy Reviews*, 3(2):100071, 2024.
- [4] Edyta Brzychczy, Milda Aleknonytė-Resch, Dominik Janssen, and Agnes Koschmider. Process mining on sensor data: a review of related works. *Knowledge and Information Systems*, 67:4915–4948, 2025.
- [5] Zhe Chen, Fu Xiao, Fangzhou Guo, and Jinyue Yan. Interpretable machine learning for building energy management: A state-of-the-art review. *Advances in Applied Energy*, 9:100123, 2023.
- [6] Eliot Crowe, Yimin Chen, Hayden Reeve, David Yuill, Amir Ebrahimifakhar, Yuxuan Chen, Lucas Troup, Amanda Smith, and Jessica Granderson. Empirical analysis of the prevalence of HVAC faults in commercial buildings. *Science and Technology for the Built Environment*, 29(10):1027–1038, 2023.
- [7] Qiao Deng, Zhiwen Chen, Wanting Zhu, Zefan Li, Yifeng Yuan, and Yalin Wang. Adaptive fusion graph convolutional network based interpretable fault diagnosis method for HVAC systems enhanced by unlabeled data. *Energy and Buildings*, 324:114901, 2024.
- [8] William Goetzler, Richard Shandross, Jim Young, Oxana Petritchenko, Decker Ringo, and Sam McClive. Energy savings potential and RD&D opportunities for commercial building HVAC systems. Report DOE/EE-1703, U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy, Building Technologies Office, December 2017. Prepared by Navigant Consulting, Inc.
- [9] Jessica Granderson, Rupam Singla, Ebony Mayhorn, Paul Ehrlich, Draguna Vrabie, and Stephen Frank. Characterization and survey of automated fault detection and diagnostic tools. Technical Report LBNL-2001075, Lawrence Berkeley National Laboratory, Berkeley, CA, November 2017.
- [10] H. Burak Gunay, Jayson Bursill, Brent Huchuk, and Scott Shillinglaw. Inverse model-based detection of programming logic faults in multiple zone VAV AHU systems. *Building and Environment*, 211:108732, 2022.
- [11] Srinivas Katipamula and Michael R. Brambley. Methods for fault detection, diagnostics, and prognostics for building systems—a review, part I. *HVAC&R Research*, 11(1):3–25, 2005.
- [12] Guannan Li, Luhan Wang, Limei Shen, Liang Chen, Hengda Cheng, Chengliang Xu, and Fan Li. Interpretation of convolutional neural network-based building HVAC fault diagnosis model using improved layer-wise relevance propagation. *Energy and Buildings*, 286:112949, 2023.
- [13] Adil Mukhtar, Michael Hadwiger, Franz Wotawa, and Gerald Schweiger. Reproducibility of machine learning-based fault detection and diagnosis for HVAC systems in buildings: An empirical study. *Energy and AI*, 22:100658, 2025. Also arXiv:2508.00880.

- [14] Tadao Murata. Petri nets: Properties, analysis and applications. *Proceedings of the IEEE*, 77(4):541–580, 1989.
- [15] Jason Priem, Heather Piwowar, and Richard Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts. *arXiv:2205.01833*, 2022.
- [16] Prince, Byungun Yoon, and Prashant Kumar. Fault detection and diagnosis in air-handling unit (AHU) using improved hybrid 1D convolutional neural network. *Systems*, 13(5):330, 2025.
- [17] Marco Pritoni, Guanqing Lin, Yimin Chen, Raphael Vitti, Christopher Weyandt, and Jessica Granderson. From fault-detection to automated fault correction: A field study. *Building and Environment*, 214:108900, 2022.
- [18] Jeffrey Schein, Steven T. Bushby, Natascha S. Castro, and John M. House. A rule-based fault detection method for air handling units. *Energy and Buildings*, 38(12):1485–1492, 2006.
- [19] Wunna Tun, Johnny Kwok-Wai Wong, and Sai-Ho Ling. Hybrid random forest and support vector machine modeling for HVAC fault detection and diagnosis. *Sensors*, 21(24):8163, 2021.
- [20] United Nations Environment Programme and Global Alliance for Buildings and Construction. Global status report for buildings and construction 2024/2025: Not just another brick in the wall. Technical report, United Nations Environment Programme, Nairobi, 2025. Buildings and construction: 32 % of global final energy use and 34 % of energy-related CO₂ emissions in 2023.
- [21] Wil M. P. van der Aalst. *Process Mining: Data Science in Action*. Springer, 2nd edition, 2016.
- [22] Hong Wang, Wenjing Yan, Yajuan Zhang, Jianqiang Yu, Wensheng Gao, Bowen Zhou, and Yi Liu. xMHAC-PDCNN: An explainable deep network with spatial feature embedding for fault detection and diagnosis of chillers under variable operating conditions. *Building and Environment*, 284:113504, 2025.
- [23] Chenglong Xiong, Guannan Li, Ying Yan, Hanyuan Zhang, Chengliang Xu, and Liang Chen. Effects of various information scenarios on layer-wise relevance propagation-based interpretable convolutional neural networks for air handling unit fault diagnosis. *Building Simulation*, 17(10):1709–1730, 2024.